

Loops Assignment

Question 1: Daily Attendance

A teacher wants to record attendance for 5 students.

Use a for loop to ask the name of each student and display:

Attendance marked for: [student name]

After all 5 students have been entered, display:

Attendance completed.

Question 2: Multiplication Table Generator

A teacher wants a program that generates a multiplication table.

Ask the user to enter a number.

Using a for loop, display its multiplication table from 1 to 10.

For example:

Enter number: 7

$7 \times 1 = 7$

$7 \times 2 = 14$

...

$7 \times 10 = 70$

Question 3: Student Marks Analyzer

A teacher wants to enter marks for 5 students.

Use a loop to:

1. Ask for each student's marks.
2. Display whether the student passed or failed.
3. Count how many students passed.
4. Count how many students failed.

At the end, display:

Total Passed: X

Total Failed: Y

Question 4: Shopping Cart Total

A customer is buying products from an online store.

The program should ask the user to enter the price of 5 products.

Use a loop to calculate the total price.

After all prices have been entered, display:

Total Shopping Amount: Rs. XXXXX

Question 5: Number Analyzer

A user wants to enter 10 numbers.

Use a for loop to analyze each number.

Your program should count:

- How many numbers are positive
- How many numbers are negative
- How many numbers are zero

At the end, display all three counts.

Example:

Positive Numbers: 4

Negative Numbers: 5

Zeroes: 1

Question 6: Password Retry System

You are developing a login system.

The correct password is:

python123

Use a while loop to repeatedly ask the user for the password.

The program should continue until the user enters the correct password.

If the password is correct:

Login Successful

If it is incorrect:

Incorrect Password. Try Again.

Question 7: ATM Menu

Create a simple ATM system using a while loop.

The initial balance is:

50000

Display this menu repeatedly:

1. Check Balance
2. Deposit
3. Withdraw
4. Exit

The program should continue until the user selects 4.

Implement the following:

- Check balance
 - Deposit money
 - Withdraw money
 - Prevent withdrawal if the amount is greater than the balance
 - Exit the program
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Question 8: Number Guessing Game

Create a number guessing game.

Set the secret number to:

secret_number = 25

Ask the user to guess the number.

Use a while loop.

If the guess is:

- Greater than 25 → "Too High"
- Less than 25 → "Too Low"
- Equal to 25 → "Correct!"

Also count how many attempts the user made.

Example:

Guess: 30

Too High

Guess: 20

Too Low

Guess: 25

Correct!

Attempts: 3

Question 9: Student Marks Entry System

A teacher wants to enter student marks continuously.

Use a while loop.

The program should keep asking for marks until the teacher enters:

-1

For every student's marks:

- 50 or above → Pass
- Below 50 → Fail

At the end, display:

Total Students

Total Passed

Total Failed

The -1 should **not** be counted as a student's marks.

Question 10: Restaurant Ordering System

You are developing a simple restaurant ordering system.

Display this menu repeatedly:

1. Burger - Rs. 500
2. Pizza - Rs. 1000
3. Fries - Rs. 300
4. Drink - Rs. 150
5. Checkout

Use a while loop.

When the customer selects an item:

- Add its price to the total bill.

- Display the selected item.
- Continue showing the menu.

When the customer selects 5, stop taking orders and display the final bill.

Example:

Burger added

Pizza added

Drink added

Total Bill = Rs. 1650

Question 11: Student Performance Management System

A teacher wants to create a system for analyzing the performance of multiple students.

First ask:

Enter number of students:

For each student, ask:

- Student name
- Marks in Python
- Marks in Mathematics
- Marks in English

Calculate:

Total Marks

Percentage

Result Rules

A student is considered **Pass** only if:

- Overall percentage is at least 50%
- AND marks in every subject are at least 40.

Otherwise, the student is **Fail**.

Grade Rules

For students who pass:

90+ → A+

80-89 → A

70-79 → B

60-69 → C

50-59 → D

While processing students, calculate:

- Number of passed students
- Number of failed students
- Highest percentage
- Lowest percentage
- Class total marks
- Class average percentage

Final Display:

===== CLASS REPORT =====

Total Students: 10

Passed: 7

Failed: 3

Highest Percentage: 94%

Lowest Percentage: 42%

Class Average: 76.5%

Important requirement

Use a **loop** to process all students.

Do not manually write the same code for every student.

Question 12: Restaurant Ordering & Billing System

You are developing a restaurant ordering system.

The restaurant has the following menu:

MENU

1. Burger Rs. 500

2. Pizza Rs. 1000

3. Biryani Rs. 350

4. Fries Rs. 250

5. Drink Rs. 150

6. Checkout

The program should repeatedly display the menu and allow the customer to order multiple items.

The program should continue until the customer selects:

6. Checkout

When an item is selected

Add its price to the total bill.

Also keep track of how many items the customer ordered.

For example:

Burger added

Pizza added

Drink added

Invalid Input

If the customer enters something other than 1-6:

Invalid choice. Please try again.

The program should **not terminate**.

Discount Rules

When the customer checks out:

Bill $\geq$ 5000

→ 20% discount

Bill $\geq$ 3000

→ 15% discount

Bill >= 1500

→ 10% discount

Below 1500

→ No discount

Delivery Charges

After discount:

Final bill >= 3000

→ Free Delivery

Final bill < 3000

→ Rs. 200 Delivery Charges

Finally display:

===== FINAL BILL =====

Total Items: 5

Original Bill: Rs. 3500

Discount: Rs. 525

Delivery Charges: Rs. 0

Final Bill: Rs. 2975

Thank you for ordering!

Important requirement

Use a **while loop** because the number of orders is unknown. The customer should be able to keep ordering until they choose checkout.

Concepts tested

- Variables
- Input
- Data types
- Arithmetic operators
- Comparison operators
- Logical conditions
- if/elif/else
- while loop

- Counters
- Accumulators
- Menu-based program
- f-strings
- Basic problem solving